

Cut Rules in DAG-Based BFT Mempools: From Relaxed Mempool to Adaptive Cuts

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1. Motivation: The Cut Rule Problem

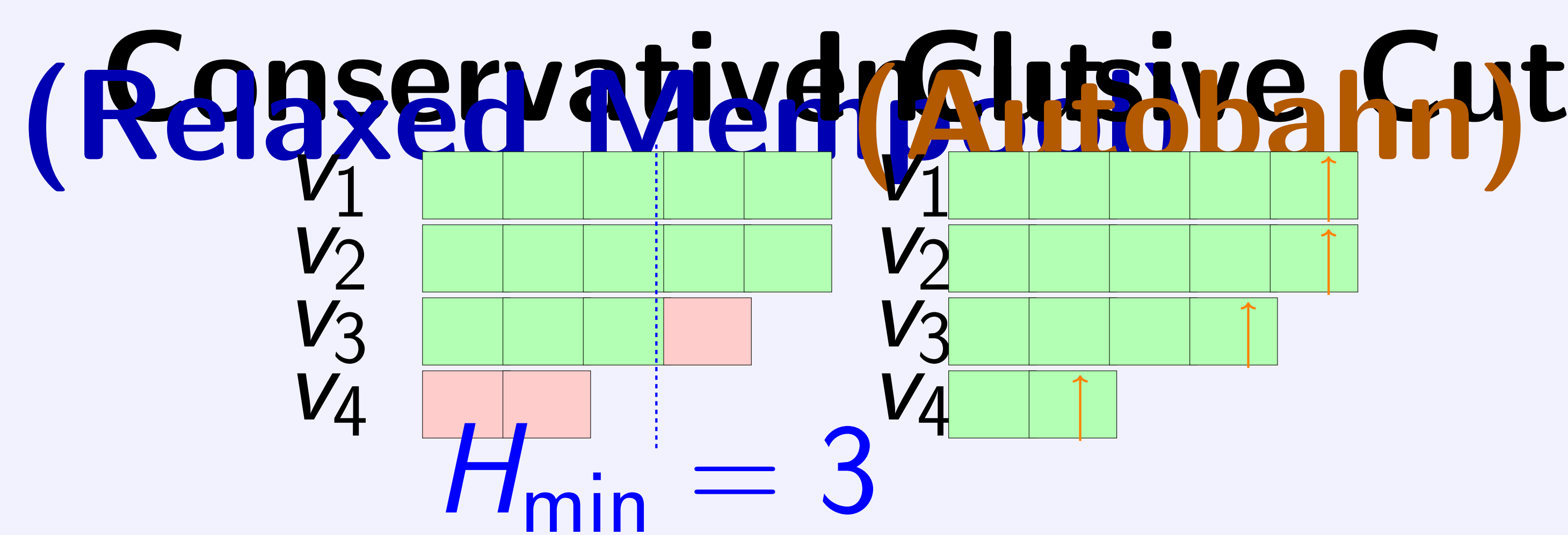
DAG-based BFT protocols decouple data dissemination from ordering. Validators broadcast batches into a DAG; a consensus leader periodically **cuts** the DAG to commit transactions.

Key Question: How should the leader decide *which* data to include in each commit?

The **cut rule** determines:

- **Straggler resilience:** Do slow validators delay commits?
- **Transaction fairness:** Are all validators' data included?
- **Hangover effect:** How fast does the system recover after blips?

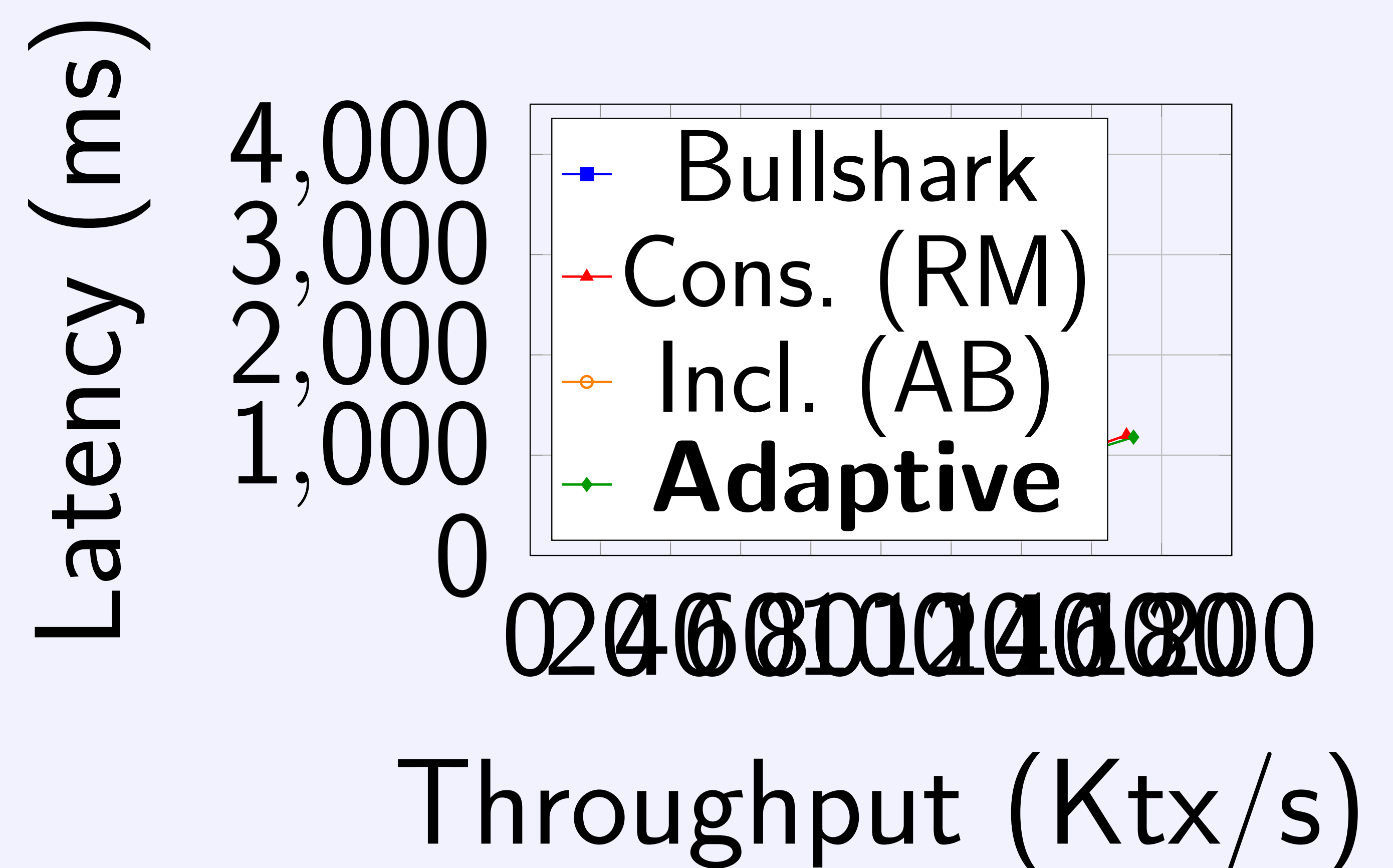
2. Two Existing Cut Strategies



4. Experimental Results

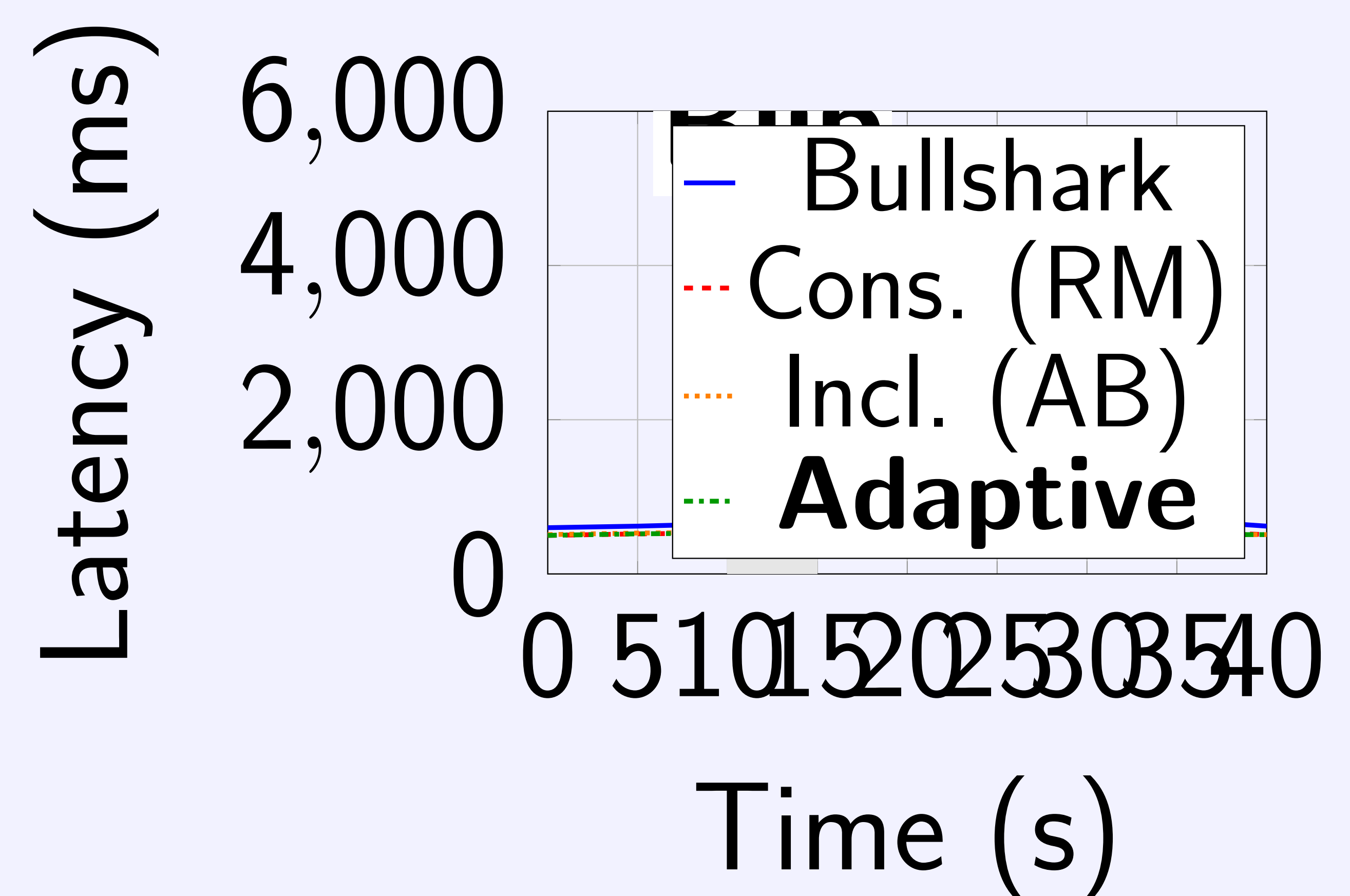
Setup: 4 servers (i9-13900K, 64 GB), 10 validators, 512 B tx, 500 KB batches.

(a) **Straggler Scenario** (4 validators with +300 ms delay)



Adaptive $\approx$ Conservative under stragglers (high Δ triggers Relaxed mode).

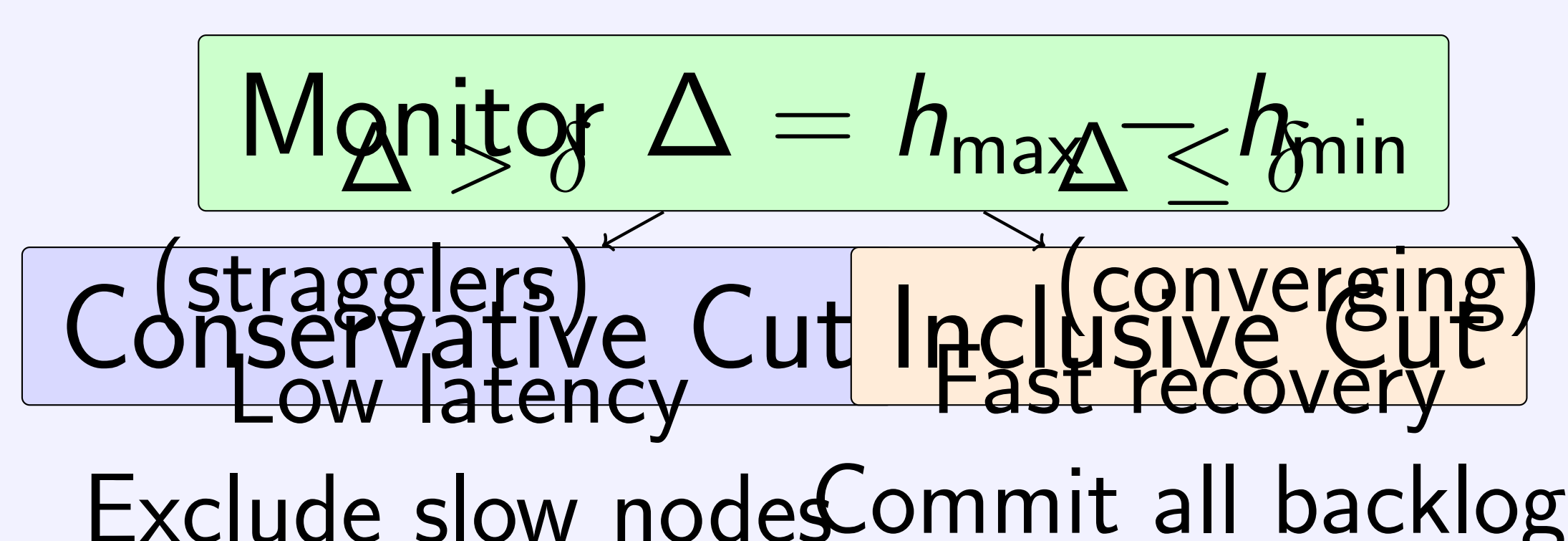
(b) **Hangover Effect** (5s network blip at $t=10s$)



Adaptive $\approx$ Inclusive during recovery (Δ drops $\rightarrow$ switches to Inclusive mode).

3. Our Proposal: Adaptive Cut

Key Insight: The two strategies have *complementary* strengths. We unify them with an **adaptive cut** that monitors *height disparity* ($\Delta = h_{\max} - h_{\min}$) across validators.



Both strategies operate on the **same DAG substrate**—switching is zero-overhead.

5. Key Takeaways

Scenario	Cons. (RM)	Incl. (AB)	Adaptive
Straggler (4 nodes)	170K tx/s	160K tx/s	172K tx/s
Straggler latency	1.2s	1.5s	1.18s
Blip recovery time	20s	2s	2.5s
Happy path	195K	205K	200K

The cut rule is a first-class design dimension in DAG-based BFT.

The adaptive cut achieves the best of both worlds by dynamically switching based on validator height disparity—**zero overhead**, same DAG substrate.

References

- [1] Danezis et al., “Narwhal and Tusk,” *EuroSys*, 2022. [2] Spiegelman et al., “Bullshark,” *ACM CCS*, 2022.
[3] Giridharan et al., “Autobahn: Seamless high speed BFT,” *SOSP*, 2024. [4]